

Shades of Success: Characterizing Manipulation Quality through Object-Centric Constraints

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Abstract—Robotic manipulation is commonly evaluated using binary success or failure labels, which provide limited information about the quality of an execution when multiple successful trajectories satisfy the same task objective with different levels of geometric precision. This limitation is particularly detrimental to data-driven robot learning where demonstrations are used both for training and evaluation, and low-quality successful executions introduce undesirable variability into datasets. To address this limitation, we introduce Shades of Success, an object-centric framework for continuously characterizing manipulation quality from successful demonstrations. Our approach learns task-dependent geometric constraints from the variability of relative object motion across demonstrations, identifying directions that must remain tightly regulated while leaving unconstrained degrees of freedom unpenalized. New executions are evaluated according to their adherence to these learned constraints throughout the task and their final goal satisfaction, producing an interpretable continuous Task Quality Index. We evaluate the framework on four tabletop manipulation tasks in ManiSkill using clean, failed, and deliberately degraded-but-successful executions. Results show that clean demonstrations receive consistently high quality scores, while successful executions containing slips, object displacement, or fumbling are assigned intermediate scores that conventional binary evaluation cannot distinguish.

I. INTRODUCTION

A successful manipulation task does not necessarily imply a high-quality execution: two successful executions can differ substantially in how precisely the object is moved, how closely geometric constraints are maintained, or how much unintended motion occurs along the way. These differences are particularly important when robotic systems learn from demonstrations, where executions that are labeled identically as “successful” may provide very different examples of desirable behavior. However, manipulation performance is often reduced to task completion, with trajectories classified simply as successful or unsuccessful. While convenient for benchmarking imitation learning policies, particularly large-scale vision-language-action (VLA) models, trained on heterogeneous demonstrations [1]–[3], this convention masks critical variations in geometric precision and execution quality. Recent analysis reveals that over half of the “successful” executions in standard datasets exhibit marginal quality or fumbling, demonstrating that binary outcomes are often uninformative

about how well a task was actually performed [4]. Because data-driven learning systems rely on these same trajectories for both training and evaluation, treating all predicate-satisfying trajectories equally injects undesirable variability that behavior-cloned policies faithfully reproduce [5].

To address this, a growing body of work attempts to filter or reweight mixed-quality data using learned expertise estimates [6], expert references [7], [8], or self-supervised transition filtering [9]; others leave the dataset intact and instead steer the policy at inference time via a learned critic [10]. However, both families reduce execution quality to an opaque scalar that cannot report which specific geometric degrees of freedom were violated. Similarly, vision-language models fine-tuned as categorical success detectors [11]–[13] remain weak judges of fine spatial tolerances. Other efforts to score continuous execution rely on robot-centric methods like kinematic smoothness [4], [14] or goal progression [15], [16]. Because these measures monitor the robot rather than the manipulated item, they fail to capture object-centric degradation, such as an object slipping within the grasp despite a smooth end-effector trajectory. Furthermore, these task-agnostic metrics ignore critical physical constraints, treating precise insertion phases and free-space transport identically.

To bridge this gap, we propose a Task Quality Index that measures geometric execution quality as adherence to task-specific kinematic constraints inferred from successful demonstrations. The underlying idea is that variability across demonstrations provides evidence of which aspects of an interaction are consistently regulated and how much deviation is admissible. We represent these interactions through relative object configurations at corresponding stages of the task. Spectral analysis of this variability identifies constrained directions and estimates their admissible variation, allowing the constraint structure to change as execution progresses.

An unseen execution is then evaluated against this learned reference. Deviations beyond the admissible variation along constrained directions incur a cost, while variation along unconstrained directions remains unpenalized. Aggregating these costs gives a scalar quality index that captures geometric departures throughout execution, including those corrected before task completion.

II. METHODOLOGY

The proposed method consists of four main steps: (1) semantic alignment of demonstrations, (2) object-centric representation of relative motion on $SE(3)$, (3) extraction of

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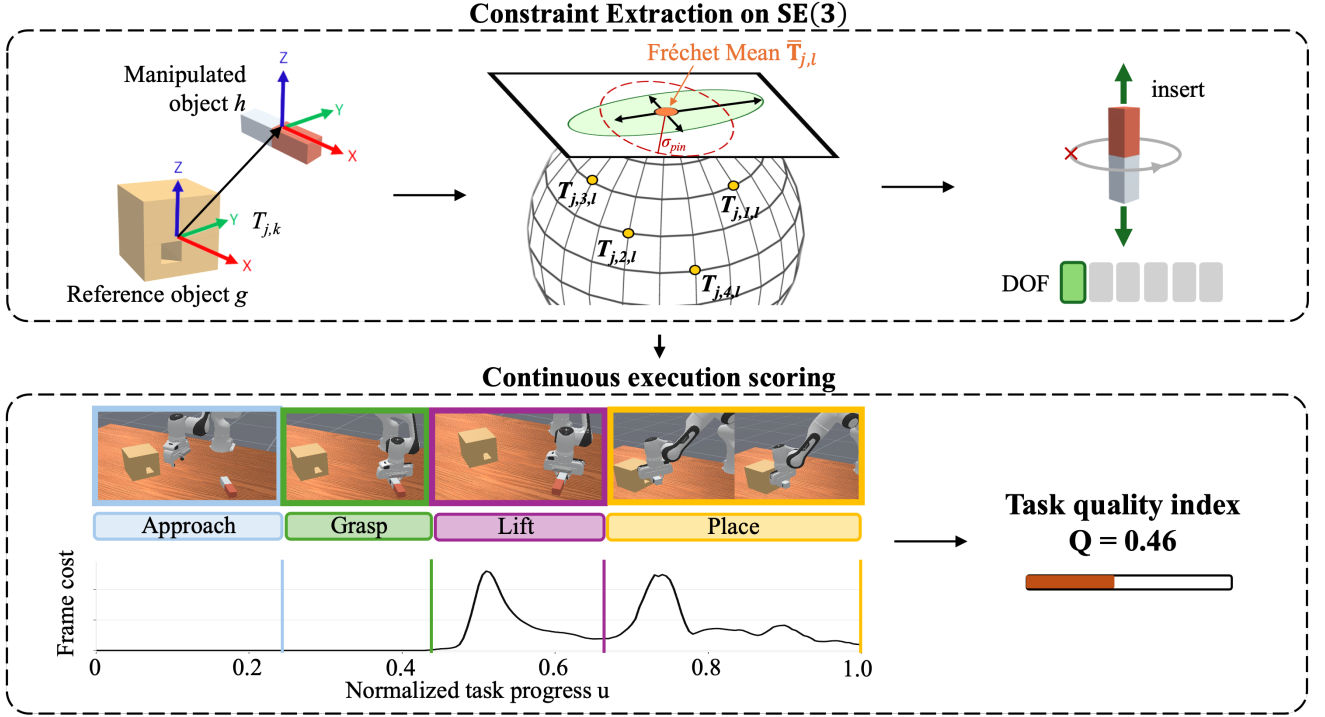


Fig. 1. Overview of the proposed framework. **Constraint Extraction:** The relative pose $\mathbf{T}_{j,k}$ between objects h and g is mapped to the $\text{SE}(3)$ tangent space. The Riemannian Fréchet mean $\bar{\mathbf{T}}_{j,l}$ defines the nominal configuration, while tangent-space covariance isolates constrained degrees of freedom using a physical threshold σ_{pin} . **Execution Scoring:** Trajectories are aligned to normalized progress u . Frame-by-frame evaluation against learned constraints yields a continuous cost sequence that localizes anomalies (e.g., slipping, fumbling). These deviations aggregate into a Task Quality Index Q , distinguishing degraded executions ($Q = 0.46$) from clean successes.

kinematic constraints in the tangent space, and (4) continuous scoring of unseen executions.

A. Semantic Alignment of Demonstrations

Let a demonstration video contain T_k unaligned frames for $k \in \{1, \dots, K\}$, where execution speed and duration vary across demonstrations. Following TREAD [17], a visual-language model first decomposes the task into P distinct semantic sub-tasks from the task instruction and scene context, and then segments each demonstration at the corresponding keyframe boundaries. Sub-task progress $u_k(t)$ is normalized to $[0, 1]$, and discretizing each phase into M progress bins yields $S = PM$ canonical progress samples across the task, mapping variable-length executions to a shared reference coordinate independent of execution speed.

B. Object-Centric Relative Motion Representation

At frame t of demonstration k , the observed pose of an object o in the scene is represented as $\mathbf{T}_{o,k}(t) = (\mathbf{R}_{o,k}(t), \mathbf{c}_{o,k}(t)) \in \text{SE}(3)$, where $\mathbf{R} \in \text{SO}(3)$ and $\mathbf{c} \in \mathbb{R}^3$ denote its orientation and position. Rather than evaluating object motion in an arbitrary global coordinate frame, we differentiate the objects in the scene into three functional categories to isolate the relative interactions that define the manipulation task. The **manipulated object** h is the active item being directly grasped and interacted with. The **reference object** g serves as the primary reference entity that defines the target configuration or trajectory for the manipulated object.

We automatically identify the reference object among multiple candidate items in the scene by finding the object whose scale-normalized translational trajectory exhibits the minimum variance across all successful demonstrations. Finally, we define as **supporting objects** s the environmental fixtures (e.g. table) and the robot’s gripper, which provide spatial bounds and workspace constraints. Each manipulated object is represented relative to the reference object, and both are additionally represented relative to the supporting objects, so that motion of the object within the grasp and displacement of the reference object itself appear as independent relations.

Let $\mathcal{C}$ denote the set of all geometrically relevant interacting object pairs in the scene. For any arbitrary pair $j = (a, b) \in \mathcal{C}$, where a serves as the local reference frame and b is the observed target object, their relative pose at frame t is computed as:

$$\mathbf{T}_{j,k}(t) = \mathbf{T}_{a,k}(t)^{-1} \mathbf{T}_{b,k}(t) \in \text{SE}(3). \quad (1)$$

To align trajectories across variable-speed demonstrations, relative poses for every pair $j \in \mathcal{C}$ are aggregated along the canonical progress coordinate. For each pair j , demonstration k , and progress sample $l \in \{0, \dots, S-1\}$, the progress-aligned pose $\mathbf{T}_{j,k,l}$ is the Lie-group mean of the frames falling within the bin, so that every demonstration carries equal weight at every sample. The nominal relative configuration is then

computed as the Riemannian Fréchet mean $\bar{\mathbf{T}}_{j,l}$:

$$\sum_{k=1}^K \text{Log}(\bar{\mathbf{T}}_{j,l}^{-1} \mathbf{T}_{j,k,l}) = \mathbf{0}. \quad (2)$$

C. Constraint Estimation and Physical Tolerances

The progress-aligned relative trajectories define a set of physically meaningful deviations around nominal behavior. To compute statistics on the curved SE(3) manifold, we map these poses to the tangent space, establish a nominal configuration, construct a physical metric, and learn directional tolerances.

1) *Tangent-Space Representation and Fréchet Mean*: We map poses to the local tangent space (the Lie algebra $\mathfrak{se}(3) \cong \mathbb{R}^6$) using the logarithmic map. The nominal relative trajectory at progress sample l is defined as the Riemannian Fréchet mean $\bar{\mathbf{T}}_{j,l}$, which minimizes the sum of squared geodesic distances to all demonstrated poses. It is found iteratively by driving the sum of local tangent-space errors to zero, as defined in (2). With the nominal pose $\bar{\mathbf{T}}_{j,l} = (\bar{\mathbf{R}}, \bar{\mathbf{c}})$ established, the relative deviation of an observed pose $\mathbf{T}_{j,k,l} = (\mathbf{R}, \mathbf{c})$ from the mean is represented as the twist:

$$\boldsymbol{\xi}_{j,k,l} = \text{Log}(\bar{\mathbf{T}}_{j,l}^{-1} \mathbf{T}_{j,k,l}) = \begin{bmatrix} \boldsymbol{\omega}_{j,k,l} \\ \mathbf{v}_{j,k,l} \end{bmatrix}, \quad (3)$$

where $\boldsymbol{\omega}_{j,k,l} = \text{Log}(\bar{\mathbf{R}}^{-1} \mathbf{R}) \in \mathfrak{so}(3) \cong \mathbb{R}^3$ represents rotation and $\mathbf{v}_{j,k,l} = \mathbf{V}(\boldsymbol{\omega}_{j,k,l})^{-1} \bar{\mathbf{R}}^{-1}(\mathbf{c} - \bar{\mathbf{c}}) \in \mathbb{R}^3$ is the translation corrected by the left SE(3) Jacobian $\mathbf{V}$.

Given a set of $|\mathcal{K}_h| = 6$ non-collinear 3D keypoints $\mathcal{K}_h \subset \mathbb{R}^3$ obtained by greedy farthest-point sampling of the manipulated object h 's rest shape, a point $\mathbf{p} \in \mathcal{K}_h$ undergoes a physical displacement $\delta \mathbf{p} = \boldsymbol{\omega} \times \mathbf{p} + \mathbf{v} = [-\hat{\mathbf{p}} \quad \mathbf{I}_3] \boldsymbol{\xi}$ under a twist deviation $\boldsymbol{\xi}$. Averaging the squared displacement over this keypoint set yields the pullback metric:

$$d_j^2(\boldsymbol{\xi}) = \frac{1}{|\mathcal{K}_h|} \sum_{\mathbf{p} \in \mathcal{K}_h} \|\boldsymbol{\omega} \times \mathbf{p} + \mathbf{v}\|_2^2 = \boldsymbol{\xi}^\top \mathbf{G}_j \boldsymbol{\xi}, \quad (4)$$

where the metric tensor $\mathbf{G}_j$ is defined as:

$$\mathbf{G}_j = \frac{1}{|\mathcal{K}_h|} \sum_{\mathbf{p} \in \mathcal{K}_h} \begin{bmatrix} -\hat{\mathbf{p}}^2 & \hat{\mathbf{p}} \\ -\hat{\mathbf{p}} & \mathbf{I}_3 \end{bmatrix}. \quad (5)$$

The symmetric matrix root $\mathbf{W}_j = \mathbf{G}_j^{1/2}$ maps twists into metric-whitened coordinates $\tilde{\boldsymbol{\xi}} = \mathbf{W}_j \boldsymbol{\xi}$, such that $\|\tilde{\boldsymbol{\xi}}\|_2^2$ directly measures mean squared physical displacement in m^2 .

2) *Constraint Spectrum*: The cross-demonstration sample covariance of the whitened deviations at progress sample l is the 6×6 matrix

$$\boldsymbol{\Sigma}_{j,l} = \frac{1}{K-1} \sum_{k=1}^K \tilde{\boldsymbol{\xi}}_{j,k,l} \tilde{\boldsymbol{\xi}}_{j,k,l}^\top = \mathbf{E}_{j,l} \boldsymbol{\Lambda}_{j,l} \mathbf{E}_{j,l}^\top, \quad (6)$$

where $\boldsymbol{\Lambda}_{j,l} = \text{diag}(\nu_{j,1}(l), \dots, \nu_{j,6}(l))$ contains the eigenvalues in descending order and the columns of $\mathbf{E}_{j,l}$ are the corresponding principal directions. A minimum standard deviation σ_0 , expressed in physical displacement units, is enforced on the spectrum, keeping the covariance positive definite for the interpolation of Sec. II-D.

3) *Constrained Directions and Degrees of Freedom*: The spectral decomposition of the metric-whitened covariance $\boldsymbol{\Sigma}_{j,l}$ separates the six local tangent directions into strictly constrained axes and free degrees of freedom (DOFs). The physical intuition is that successful demonstrations will naturally exhibit high variability in unconstrained task directions, whereas geometric contact with the environment forces near-zero variance along constrained axes.

To classify these states, the spread along each principal axis i is read directly as a physical length,

$$\sigma_{j,i}(l) = \sqrt{\nu_{j,i}(l)}, \quad i \in \{1, \dots, 6\}, \quad (7)$$

in metres, and compared against a fixed length $\sigma_{\text{pin}} > \sigma_0$.

A direction held to less than σ_{pin} indicates a consistently regulated direction of relative motion, while a wider one identifies an unconstrained DOF. The constrained index set $\mathcal{I}_j(l)$, the instantaneous constraint codimension $n_j(l)$, and the number of remaining free DOFs $d_{\text{free},j}(l)$ are then

$$\mathcal{I}_j(l) = \{i : \sigma_{j,i}(l) < \sigma_{\text{pin}}\}. \quad (8)$$

This instantaneous rank mapping allows the framework to extract continuous changes in constraint structure along the trajectory without manual annotation.

D. Continuous Execution Validation and Task Scoring

To evaluate an unseen execution, each frame is graded continuously against the learned constraint model, culminating in an aggregate trajectory quality score.

1) *Continuous Reference Interpolation*: For a query frame at time t , the visual-language model maps t to the canonical progress coordinate $u(t)$. The reference is interpolated between adjacent discrete Fréchet means on $\text{SO}(3) \times \mathbb{R}^3$, using Slerp in rotation and linear interpolation in translation, and the regularized covariances Log-Euclidean. This yields a continuous nominal reference $\bar{\mathbf{T}}_j(u(t))$ with principal directions $\mathbf{E}_{j,i}(t)$ and spreads $\sigma_{j,i}(t)$, and hence the active set $\mathcal{I}_j(t)$ by (8). The instantaneous twist deviation is then computed as:

$$\boldsymbol{\xi}_j(t) = \text{Log}(\bar{\mathbf{T}}_j(u(t))^{-1} \mathbf{T}_j(t)), \quad (9)$$

with whitened form $\tilde{\boldsymbol{\xi}}_j(t) = \mathbf{W}_j \boldsymbol{\xi}_j(t)$.

2) *Reduced Degrees-of-Freedom Frame Cost*: At each time step, multiple geometric constraints $j \in \mathcal{C}(t)$ may be active. Deviation is evaluated only along their constrained directions, each normalized by its learned tolerance,

$$z_{j,i}(t) = \frac{\mathbf{E}_{j,i}(t)^\top \tilde{\boldsymbol{\xi}}_j(t)}{\max\{\sigma_{j,i}(t), \sigma_0\}}, \quad i \in \mathcal{I}_j(t), \quad (10)$$

so that variation along unconstrained directions remains unpunished. Directions with equal tolerance form groups $m \in \mathcal{G}_j(t)$ with coordinate vectors $\mathbf{z}_{j,m}(t)$, scored by their norm. Applying the same normalization to the demonstrations defines the task-level admissible radius z_0 , the q -th quantile of the demonstrated group norms pooled over all pairs, demonstrations, and progress samples. Because the number of constrained dimensions $n_j(t)$ varies along the trajectory, raw

deviation sums would disproportionately penalize constrained phases. We therefore normalize by the total active direction count, formulating the frame cost $c(t)$ as:

$$c(t) = \frac{\sum_{j \in \mathcal{C}(t)} \sum_{m \in \mathcal{G}_j(t)} \max \{ \|z_{j,m}(t)\|_2 - z_0, 0 \}}{\sum_{j \in \mathcal{C}(t)} n_j(t)}, \quad (11)$$

where only deviations beyond z_0 contribute. Frames with no active direction are excluded from aggregation.

3) *Terminal Gating and Quality Index*: The total execution quality combines continuous path adherence with terminal goal satisfaction. The path cost is the mean frame penalty over all active frames,

$$c_{\text{path}} = \frac{1}{|\mathcal{T}_{\text{act}}|} \sum_{t \in \mathcal{T}_{\text{act}}} c(t), \quad (12)$$

while terminal satisfaction is assessed through the final-frame cost $c_{\text{fin}} = c(T)$, with the terminal gate satisfied when $c_{\text{fin}} \leq 1$. The aggregate Task Quality Index $Q \in [0, 1]$ is defined as:

$$Q = \exp \left(-\frac{1}{2} c_{\text{path}} \right) \mathbb{I}[c_{\text{fin}} \leq 1]. \quad (13)$$

A score of $Q = 1$ indicates that all evaluated residuals remain within the learned admissible radius; lower values indicate greater average deviation beyond it along the constrained directions, and $Q = 0$ when the terminal gate fails.

III. EXPERIMENTS

While binary labels indicate task completion, our proposed quality index Q captures continuous deviations from nominal behavior. We assess its ability to cleanly separate clean demonstrations from degraded successes.

Setup. We evaluate four ManiSkill [18] tabletop tasks spanning varied contact states: *StackCube*, *PlaceSphere*, *PegInsertionSide*, and *PlacePen*. For each task, a constraint model is learned from 30 clean demonstrations. To validate our framework, we create an evaluation dataset comprising 20 trajectories per category across three distinct performance tiers: successful executions, outright task failures where terminal constraints are unmet, and four dedicated fault suites designed to test degraded successful executions. Specifically, the *slip* fault introduces minor object slippage within the gripper during transit; the *nudge* fault involves the robot inadvertently displacing the reference object by 5 cm prior to placement; *slip + nudge* combines both anomalies sequentially; and *fumble* captures trajectories where the robot drops and subsequently recovers the object mid-task while still successfully achieving the terminal goal.

Parameters. Each semantic phase is discretized into $M = 12$ progress bins. The constraint threshold is fixed at $\sigma_{\text{pin}} = 5$ mm, the minimum spread at $\sigma_0 = 1$ mm, and the admissible-radius quantile at $q = 0.95$.

IV. PRELIMINARY RESULTS AND DISCUSSION

As detailed in Table I, our initial simulation results demonstrate that the Quality Index (Q) reliably stratifies execution fidelity and captures a gradual transition in task performance. Clean successful trajectories consistently achieve high scores ($Q \geq 0.97$), whereas genuine task failures collapse to near zero due to severe terminal penalties (c_{fin}). Crucially, the introduced fault suites exhibit intermediate quality scores and elevated path costs (c_{path}). This decomposition localizes geometric anomalies: c_{path} grades *how* the goal was reached, with Q decaying exponentially in it, while c_{fin} decides *whether* it was reached, entering only as a gate that zeroes Q when $c_{\text{fin}} > 1$.

However, sensitivity to degradation varies by task. For example, *Slip* separates cleanly from success in *StackCube* (0.19 vs 1.00), but is barely distinguishable in *PlacePen* (0.93 vs 0.97), where *nudge* is also largely missed ($c_{\text{path}} = 0.12$, against 1.02–1.63 in the other three tasks). Two properties of the learned model account for this. First, the clean demonstrations of *PlacePen* are themselves the most variable ($Q = 0.97 \pm 0.11$), which widens the learned tolerances $\sigma_{j,i}$ and the admissible radius z_0 , so moderate deviations fall below the penalty threshold in (11). Second, the pullback metric (5) is built from keypoints on the manipulated object, so for a thin object a given in-hand rotation maps to a far smaller physical displacement than the same rotation of a cube. Only gross events then exceed z_0 , which is why *fumble* remains the most penalized suite for this task ($Q = 0.35$, $c_{\text{path}} = 2.19$).

TABLE I
QUALITY INDEX AND COSTS PER POPULATION ACROSS TASKS.

	Q	c_{path}	c_{fin}		Q	c_{path}	c_{fin}
<i>StackCube</i>				<i>PlaceSphere</i>			
success	1.00 $\pm$.00	0.01	0.07	success	1.00 $\pm$.00	0.01	0.04
slip	0.19 $\pm$.39	0.38	0.12	slip	0.78 $\pm$.40	0.14	0.10
nudge	0.53 $\pm$.04	1.28	0.07	nudge	0.50 $\pm$.45	1.02	0.19
slip + nudge	0.09 $\pm$.20	1.48	0.15	slip + nudge	0.19 $\pm$.34	0.93	0.35
fumble	0.85 $\pm$.02	0.33	0.10	fumble	0.71 $\pm$.16	0.66	0.07
failure	0.00 $\pm$.00	1.94	19	failure	0.00 $\pm$.00	4.29	136
<i>PegInsertionSide</i>				<i>PlacePen</i>			
success	0.99 $\pm$.01	0.01	0.03	success	0.97 $\pm$.11	0.09	0.08
slip	0.72 $\pm$.37	0.23	0.05	slip	0.93 $\pm$.12	0.20	0.21
nudge	0.43 $\pm$.13	1.63	0.06	nudge	0.91 $\pm$.19	0.12	0.14
slip + nudge	0.25 $\pm$.32	1.21	0.06	slip + nudge	0.82 $\pm$.28	0.26	0.41
fumble	0.81 $\pm$.10	0.44	0.03	fumble	0.35 $\pm$.13	2.19	0.13
failure	0.00 $\pm$.00	6.81	65	failure	0.00 $\pm$.00	5.91	220

When demonstrations share provenance, as ours do, policy-specific regularities such as a preferred approach axis may register as constraints and penalize deviation where none is physically consequential. Its reliance on SE(3) poses restricts its application strictly to rigid bodies, excluding deformable or articulated objects. Additionally, the accuracy of Q depends heavily on the temporal consistency of the vision-language model; delayed semantic segmentation can misalign the canonical progress coordinate and artificially inflate path costs.

Future work will prioritize comprehensive ablation studies on key parameters (e.g., K , M , σ_{pin} , and $\mathcal{C}$) across a broader suite of rigid-body manipulation tasks. Ultimately, we aim to evaluate our framework on large robot datasets.

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